

Shuran Song | Curriculum Vitae

Department of Electrical Engineering – Stanford University – Stanford, CA 94305

✉ shuran@stanford.edu • Personal Website · Google Scholar

Education

Princeton University

Ph.D. in Computer Science

2013-2018

Hong Kong University of Science and Technology

B.Eng. in Computer Engineering

2009-2013

Employment history

Stanford University

Assistant Professor

2023-now

Columbia University

Adjunct Associate Professor

2023-now

Columbia University

Assistant Professor

2019-2023

Samsung AI Center

Research Consultant

2021-2023

Google Brain

Student Researcher

2018-2019

Awards and Honors

Faculty Research Awards.....

2023: Gabilan Faculty Fellow

2022: NSF CAREER Award

2022: Sloan Research Fellows

2022: Google Research Scholar Award

2021: Microsoft Research Faculty Fellows

2021: JP Morgan Faculty Research Award

20/21/22: Amazon Research Award

2021: TRI Young Faculty Researcher Award

2020: Columbia Junior Faculty Award

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Paper Awards.....

2022 RSS : Best Paper Award, Best Student Paper Finalist, Iterative Residual Policy for Goal-Conditioned Dynamic Manipulation of Deformable Objects

2022 RSS : Best System Paper Finalist, DextAIRity: Deformable Manipulation Can be a Breeze

2021 CoRL: Best System Paper, FlingBot: The Unreasonable Effectiveness of Dynamic Manipulations for Cloth Unfolding

2021 Japan Factory Automation: Foundation Paper Award What are the important technologies for bin picking? Technology analysis of robots in competitions based on a set of performance metrics

2020 TRO: Best Paper Award, TossingBot: Learning to Throw Arbitrary Objects with Residual Physics

2020 ICRA: Best Paper in Automation Finalist, Form2Fit: Learning Shape Priors for Generalizable Assembly from Disassembly

2019 RSS: Best System Paper, TossingBot: Learning to Throw Arbitrary Objects with Residual Physics

2019 CVPR: Best Paper Finalist, Neural Illumination: Lighting Prediction for Indoor Environments

2018 IROS: Best Cognitive Robotics Paper Award Finalist, Learning Synergies between Pushing and Grasping with Self-supervised Deep Reinforcement

2018 Amazon: Best Systems Paper Award in Manipulation, Robotic Pick-And-Place of Novel Objects in Clutter with Multi-Affordance Grasping and Cross-Domain Image Matching

2018 SGP: Best Dataset Award, ShapeNet: An Information-Rich 3D Model Repository

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Awards as Students.....

2017: Wallace Fellowship

2017: Princeton School of Engineering and Applied Science Award for Excellence

2017: 1st Place Winners (Stow Task) at the worldwide Amazon Robotics Challenge 2017

2016: Siebel Scholar, Class of 2017

2016: 3rd and 4th Place Winners at the worldwide Amazon Picking Challenge 2016

2016: Qualcomm Innovation Fellowship Finalists

2015: Facebook Ph.D. Fellowship

Peer-reviewed Publications

Huy Ha, Pete Florence, Shuran Song

Scaling Up and Distilling Down: Language-Guided Robot Skill Acquisition

Conference on Robot Learning (CoRL) 2023 [Project Webpage](#)

Zeyi Liu*, Arpit Bahety*, Shuran Song

REFLECT: Summarizing Robot Experiences for Failure Explanation and Correction

Conference on Robot Learning (CoRL) 2023 [Project Webpage](#)

Mengda Xu, Zhenjia Xu, Cheng Chi, Manuela Veloso, Shuran Song

XSkill: Cross Embodiment Skill Discovery

Conference on Robot Learning (CoRL) 2023 [Project Webpage](#)

Yulong Li, Andy Zeng, Shuran Song

Rearrangement Planning for General Part Assembly

Conference on Robot Learning (CoRL) 2023 [Project Webpage](#)

Jimmy Wu, Rika Antonova, Adam Kan, Marion Lepert, Andy Zeng, Shuran Song, Jeannette Bohg, Szymon Rusinkiewicz, Thomas Funkhouser

TidyBot: Personalized Robot Assistance with Large Language Models

Autonomous Robots Special Issue: Large Language Models in Robotics, (AuRo) 2023

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023 [Project Webpage](#)

Neil Nie, Samir Yitzhak Gadre, Kiana Ehsani, Shuran Song

Structure From Action: Learning Interactions for Articulated Object 3D Structure Discovery

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023 [Project Webpage](#)

Arpit Bahety*, Shreeya Jain*, Huy Ha, Nathalie Hager, Benjamin Burchfiel, Eric Cousineau, Siyuan Feng, Shuran Song

Bag All You Need: Learning a Generalizable Bagging Strategy for Heterogeneous Objects

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2023 [Project Webpage](#)

Cheng Chi, Siyuan Feng, Yilun Du, Zhenjia Xu, Eric Cousineau, Benjamin Burchfiel, Shuran Song

Diffusion Policy: Visuomotor Policy Learning via Action Diffusion

Robotics: Science and Systems (RSS) 2023 [Project Webpage](#)

Zhenjia Xu, Zhou Xian, Xingyu Lin, Cheng Chi, Zhiao Huang, Chuang Gan, Shuran Song

RoboNinja: Learning an Adaptive Cutting Policy for Multi-Material Objects

Robotics: Science and Systems (RSS) 2023 [Project Webpage](#)

Alper Canberk, Cheng Chi, Huy Ha, Benjamin Burchfiel, Eric Cousineau, Siyuan Feng, Shuran Song
Cloth Funnels: Canonicalized-Alignment for Multi-Purpose Garment Manipulation
International Conference on Robotics and Automation (ICRA) 2023 [📄 Project Webpage](#)

Jingxi Xu*, Han Lin*, Shuran Song, Matei Ciocarlie
TANDEM3D: Active Tactile Exploration for 3D Object Recognition
International Conference on Robotics and Automation (ICRA) 2023 [📄 Project Webpage](#)

Huy Ha, Shuran Song
Semantic Abstraction: Open-World 3D Scene Understanding from 2D Vision-Language Models
Conference on Robot Learning (CoRL) 2022 [📄 Project Webpage](#)

Zeyi Liu, Zhenjia Xu, Shuran Song
BusyBot: Learning to Interact, Reason, and Plan in a BusyBoard Environment
Conference on Robot Learning (CoRL) 2022 [📄 Project Webpage](#)

Mengda Xu, Manuela Veloso, Shuran Song
ASPiRe: Adaptive Skill Priors for Reinforcement Learning
Neural Information Processing Systems, **NeurIPS** 2022 [📄 Project Webpage](#)

Gabriel Ilharco*, Mitchell Wortsman*, Samir Yitzhak Gadre*, Shuran Song, Hannaneh Hajishirzi Simon Kornblith, Ali Farhadi, Ludwig Schmidt
Patching open-vocabulary models by interpolating weights
Neural Information Processing Systems, **NeurIPS** 2022 [📄 Project Webpage](#)

Cheng Chi, Benjamin Burchfiel, Eric Cousineau, Siyuan Feng, Shuran Song
Iterative Residual Policy for Goal-Conditioned Dynamic Manipulation of Deformable Objects
Robotics: Science and Systems (RSS) 2022
The International Journal of Robotics Research (IJRR) 2023
Best Paper Award. Best Student Paper Finalist. [📄 Project Webpage](#)

Zhenjia Xu, Cheng Chi, Benjamin Burchfiel, Eric Cousineau, Siyuan Feng, Shuran Song
DextAIRity: Deformable Manipulation Can be a Breeze
Robotics: Science and Systems (RSS) 2022
Best System Paper Finalist [📄 Project Webpage](#)

Jimmy Wu, Xingyuan Sun, Andy Zeng, Shuran Song, Szymon Rusinkiewicz, Thomas Funkhouser
Learning Pneumatic Non-Prehensile Manipulation with a Mobile Blower
Robotics and Automation Letters (RA-L), 2022
Intelligent Robots and Systems (IROS), 2022 [📄 Project Webpage](#)

Jingxi Xu, Shuran Song, Matei Ciocarlie
TANDEM: Learning Joint Exploration and Decision Making with Tactile Sensors
Robotics and Automation Letters (RA-L), 2022
Intelligent Robots and Systems (IROS), 2022 [📄 Project Webpage](#)

Shubham Agrawal*, Yulong Li*, Jen-Shuo Liu, Steven K. Feiner, Shuran Song
Scene Editing as Teleoperation: A Case Study in 6DoF Kit Assembly
Intelligent Robots and Systems (IROS), 2022 [📄 Project Webpage](#)

Samir Yitzhak Gadre, Kiana Ehsani, Shuran Song, Roozbeh Mottaghi
Continuous Scene Representations for Embodied AI
IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2022

Wenji Liu, Kai Bai, Xuming He, Shuran Song, Changxi Zheng, and Xiaopei Liu
FishGym: A High-Performance Physics-based Simulation Framework for Underwater Robot Learning
International Conference on Robotics and Automation (ICRA), 2022 [📄 Project Webpage](#)

Zhenjia Xu, Zhanpeng He, Shuran Song

UMPNet: Universal Manipulation Policy Network for Articulated Objects
Robotics and Automation Letters (**RA-L**), 2022
International Conference on Robotics and Automation (**ICRA**), 2022 [📄](#) [Project Webpage](#)

Xiaolong Li, Yijia Weng, Li Yi, Leonidas Guibas, A. Lynn Abbott, Shuran Song, He Wang
Leveraging SE(3) Equivariance for Self-Supervised Category-Level Object Pose Estimation
Neural Information Processing Systems, **NeurIPS** 2021 [📄](#) [Project Webpage](#)

Huy Ha, Shuran Song
FlingBot: The Unreasonable Effectiveness of Dynamic Manipulations for Cloth Unfolding
Conference on Robot Learning (**CoRL** 2021)
Best System Paper [📄](#) [Project Webpage](#)

Cheng Chi, Shuran Song
GarmentNets: Category-Level Pose Estimation for Garments via Canonical Space Shape Completion
IEEE International Conference on Computer Vision (**ICCV**), 2021, [📄](#) [Project Webpage](#)

Samir Yitzhak Gadre, Kiana Ehsani, Shuran Song
Act the Part: Learning Interaction Strategies for Articulated Object Part Discovery
IEEE International Conference on Computer Vision (**ICCV**), 2021, [📄](#) [Project Webpage](#)

Iretiayo Akinola*, Jingxi Xu*, Shuran Song, and Peter Allen
Dynamic Grasping with Reachability and Motion Awareness
International Conference on Intelligent Robots and Systems (IROS) 2021 [📄](#) [Project Webpage](#)

Zhenjia Xu, Beichun Qi, Shubham Agrawal, Shuran Song
AdaGrasp: Learning an Adaptive Gripper-Aware Grasping Policy
International Conference on Robotics and Automation (**ICRA**), 2021 [📄](#) [Project Webpage](#)

Jimmy Wu, Xingyuan Sun, Andy Zeng, Shuran Song, Szymon Rusinkiewicz, Thomas Funkhouser
Spatial Intention Maps for Multi-Agent Mobile Manipulation
International Conference on Robotics and Automation (**ICRA**), 2021 [📄](#) [Project Webpage](#)

Boyuan Chen, Yuhang Hu, Robert Kwiatkowski, Shuran Song, Hod Lipson
Visual Perspective Taking for Opponent Behavior Modeling
International Conference on Robotics and Automation (**ICRA**), 2021 [📄](#) [Project Webpage](#)

Yiqing Liang, Boyuan Chen, Shuran Song
SSCNav: Confidence-Aware Semantic Scene Completion for Visual Semantic Navigation
International Conference on Robotics and Automation (**ICRA**), 2021 [📄](#) [Project Webpage](#)

Zhenjia Xu*, Zhanpeng He*, Jiajun Wu, Shuran Song
Learning 3D Dynamic Scene Representations for Robot Manipulation
Conference on Robot Learning (**CoRL**), 2020 [📄](#) [Project Webpage](#)

Huy Ha*, Shubham Agrawal*, Shuran Song
Fit2Form: 3D Generative Model for Robot Gripper Form Design
Conference on Robot Learning (**CoRL**), 2020 [📄](#) [Project Webpage](#)

Huy Ha, Jingxi Xu, Shuran Song
Learning a Decentralized Multi-arm Motion Planner
Conference on Robot Learning (**CoRL**), 2020 [📄](#) [Project Webpage](#)

Sebastian Hofer, Kostas E. Bekris, Ankur Handa, Juan Camilo Gamboa, Melissa Mozifian, Florian Golemo, Christopher G. Atkeson, Dieter Fox, Ken Goldberg, John Leonard, C. Karen Liu, Jan Peters, Shuran Song, Peter Welinder, Martha White.
Sim2Real in Robotics and Automation: Applications and Challenges.
IEEE Transactions on Automation Science and Engineering 2020.

Chengzhi Mao, Amogh Gupta, Vikram Nitin, Baishakhi Ray, Shuran Song, Junfeng Yang, Carl Vondrick
Multi-task Learning Increases Adversarial Robustness
European Conference on Computer Vision (**ECCV**), 2020 [📄](#) Paper

Shuran Song, Andy Zeng, Johnny Lee, Thomas Funkhouser
Grasping in the Wild: Learning 6DoF Closed-Loop Grasping from Low-Cost Demonstrations
Robotics and Automation Letters (**RA-L**) 2020
International Conference on Intelligent Robots and Systems(**IROS**) 2020 [📄](#) Project Webpage

Jimmy Wu, Xingyuan Sun, Andy Zeng, Shuran Song, Johnny Lee, Szymon Rusinkiewicz, Thomas Funkhouser
Spatial Action Maps for Mobile Manipulation
Robotics: Science and Systems (**RSS**), 2020 [📄](#) Project Webpage

Xiaolong Li, He Wang, Li Yi, Leonidas Guibas, A. Lynn Abbott, Shuran Song
Category-Level Articulated Object Pose Estimation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2020
Oral presentation [📄](#) Project Webpage

Shreeyak S. Sajjan, Matthew Moore, Mike Pan, Ganesh Nagaraja, Johnny Lee, Andy Zeng, Shuran Song
ClearGrasp: 3D Shape Estimation of Transparent Objects for Manipulation
International Conference on Robotics and Automation, (**ICRA**), 2020 [📄](#) Project Webpage

Kevin Zakka, Andy Zeng, Johnny Lee, Shuran Song
Form2Fit: Learning Shape Priors for Generalizable Assembly from Disassembly
International Conference on Robotics and Automation, (**ICRA**), 2020
Best Paper in Automation Finalist [📄](#) Project Webpage

Yen-Chen Lin, Andy Zeng, Shuran Song, Phillip Isola, TY Lin
Learning to See before Learning to Act: Visual Pre-training for Manipulation
International Conference on Robotics and Automation, (**ICRA**), 2020 [📄](#) Project Webpage

Andy Zeng, Shuran Song, Johnny Lee, Alberto Rodriguez, Thomas Funkhouser
TossingBot: Learning to Throw Arbitrary Objects with Residual Physics
IEEE Transactions on Robotics (**TRO**), 2020
Robotics: Science and Systems (**RSS**), 2019
Best System Paper [📄](#) Project Webpage

Zhenjia Xu, Jiajun Wu, Andy Zeng, Joshua Tenenbaum, **Shuran Song**
DensePhysNet: Learning Dense Physical Object Representations via Multi-step Dynamic Interactions
Robotics: Science and Systems (**RSS**), 2019 [📄](#) Project Webpage

He Wang, Srinath Sridhar, Jingwei Huang, Julien Valentin, **Shuran Song**, Leonidas J. Guibas
Normalized Object Coordinate Space for Category-Level 6D Object Pose and Size Estimation
IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**) 2019
Oral presentation [📄](#) Project Webpage

Research Grants

Individual share 4.5M+ (total 7M+)

NSF: CAREER: Active Scene Understanding By and For Robot Manipulation (PI, \$600,000)

NSF: NRI: Hierarchical Representation Learning for Robot Assistants (PI, \$500,000 = 33% total)

NSF: FMRG: Adaptable and Scalable Robot Teleoperation for Human-in-the-Loop Assembly (PI, \$2M =50% of total)

NSF: FMNet: NYC Future Manufacturing Collective (co-PI, \$15,623 = 3% total)

TRI: Learning Sequential Interactions for Dynamic Cloth Manipulation (PI \$750,000)

Sloan Research Fellowship: (PI \$75,000)

Microsoft Research Fellowship: (PI \$200,000)

JP Morgan Research Award: (PI \$75,000)
Amazon Robotics Gift: (PI \$120,000)
Amazon Research Award: (PI \$80,000)
Google Research Scholar : (PI \$60,000)
Google Research Award Gift : (PI \$200,000)

Scientific Leadership

Associate Editor: IEEE Transactions on Robotics (T-RO)
Associate Editor: The International Journal of Robotics Research (IJRR)
Diversity, Equity, and Inclusion (DEI) Chairs: CVPR 2022, 2024
Workshop Chair: Robotics: Science and System (RSS) 2022
Co-director: AI4All in Columbia University, 2022
Mentor, Columbia University and Amazon Summer Undergraduate Research Experience (SURE) Program 2022
Program committee: Grace Hopper Celebration of Women
Area Chair: CoRL 2022,2023, CVPR 2020,2021
Program Committee: Siggraph Asia 2020, Siggraph 2021
Mentor: Women in Machine Learning Workshop (WiML) in NeurIPS.
Mentor: Women in Computer Vision Workshop (WiCV) in CVPR.
Conference Reviewer: CoRL, RSS, IROS, ICRA, ECCV, ICCV, CVPR, 3DV, CVIU, IVC, NIPS, PAMI, IJCV, Siggraph, Siggraph Asia, IJRR, Multimedia, Neurocomputing

Invited Talks and Lectures

2023: Invited talk at KAIST AI Colloquium
2023: RI Seminar at Carnegie Mellon University
2023: Keynote Talk at Closing the Loop between the Vision and Language (CLVL) workshop ICCV
2023: Invited talk at Articulate Robots: Utilizing Language for Robot Learning Workshop RSS
2023: Invited talk at Robot Representations for Scene Understanding, Reasoning and Planning Workshop RSS
2023: Invited talk at Experiment-oriented Locomotion and Manipulation Research Workshop RSS
2023: Invited talk at MIT EECS Colloquium
2023: Invited talk at Stanford University Data Science Institute
2023: Invited talk at University Washington Robotics & CS Seminar Series
2023: Invited talk at UIUC Robots Seminar Series
2022: Invited talk at University of Michigan Robotics Department Seminar
2023: Invited talk at CITRIS People and Robots Seminar Series
2022: Invited talk: Sim-to-Real Robot Learning: Locomotion and Beyond. CoRL workshop. New Zealand
2022: Invited talk at Geometry, Physics, and Human Knowledge as Inductive Bias in Robot Learning. CoRL workshop. New Zealand
2022: Invited talk at Implicit Neural Representation Workshop, ICRA 2022. Pennsylvania.
2022: Invited talk at Learning 3D Representations for Shape and Appearance Workshop. Virtual.
2022: Invited talk at University of Toronto Robotics Institute, 2022. Virtual
2022: Invited talk at Stanford University Vision Seminar, 2022. Virtual
2022: Invited talk at University of Pennsylvania GRASP Lab, 2022. Virtual
2022: Invited talk at LEAP Seminar in CMU, 2022. Virtual
2022: Invited talk at Deformable Object Manipulation Workshop, ICRA 2022. Pennsylvania.
2022: Invited talk at IEEE Workshop on Reliable and Resilient Digital Manufacturing NYU, 2021. Virtual
2022: Invited talk at Combining Learning and Motion Planning for Robotics at IROS, 2021. Virtual
2021: Invited talk at TRI Research Seminar, 2021. Virtual

2021: Invited talk at Embodied AI seminar in MIT, 2021. Virtual
2021: Invited talk at Amazon robotics research, 2021. Virtual
2021: Invited talk at Allen Institute Embodied Intelligence Seminar, 2021. Virtual
2021: Invited talk at Visual Representation Learning for robotics Workshop at RSS, 2021. Virtual
2021: Invited talk at 2nd Workshop on Closing the Reality Gap in Sim2Real Transfer for Robotics, CVPR, 2020. Virtual
2021: Invited talk at 3D Scene Understanding for Vision, Graphics, and Robotics in CVPR, 2020. Virtual
2020: Learning visual representation for generalizable manipulation. *3D Scene Understanding for Vision, Graphics, and Robotics in CVPR 2020*
2020: Grasping in the Wild, *CVPR Area Chair Workshop, UCSD*
2020: Active Scene Understanding with Robot Interaction, *Rutgers Center for Cognitive Science*